

I.

$$X = \text{Spec}(R) \quad G \curvearrowright X \leadsto G \curvearrowright R \quad (g \cdot f)(x) = f(g^{-1} \cdot x) \quad f \in R$$

• The set of G -orbits X/G

• R^G

Note: $\forall f \in R^G$ gives a well-defined function on X/G

Q: $X/G \stackrel{?}{=} \text{Spec}(R^G)$? not always hold

Lemma: $G \curvearrowright X = \text{Spec}(R)$ s.t. R^G is a f.g. $\mathbb{C}$ -algebra. Let

$\pi: X \rightarrow Y = \text{Spec}(R^G)$ induced by $R^G \hookrightarrow R$. Then

(1) $\forall \phi: X \rightarrow Z$, Z affine s.t. $\phi(g \cdot x) = \phi(x) \quad \forall g \in G$,

$x \in X$ can factor through $\pi: X \rightarrow Y$.

i.e. $\exists \bar{\phi}$

$$\begin{array}{ccc} X & \xrightarrow{\phi} & Z \\ \pi \searrow & \bar{\phi} \nearrow & \\ & Y & \end{array}$$

II. Good categorical quotient

DEFINITION.

Definition 5.0.5. Let G act on X and let $\pi: X \rightarrow Y$ be a morphism that is constant on G -orbits. Then π is a **good categorical quotient** if:

(a) If $U \subseteq Y$ is open, then the natural map $\mathcal{O}_Y(U) \rightarrow \mathcal{O}_X(\pi^{-1}(U))$ induces an isomorphism

$$\mathcal{O}_Y(U) \simeq \mathcal{O}_X(\pi^{-1}(U))^G.$$

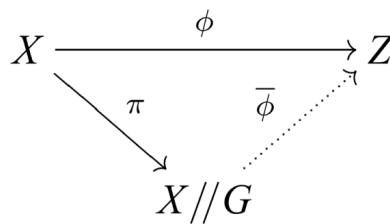
(b) If $W \subseteq X$ is closed and G -invariant, then $\pi(W) \subseteq Y$ is closed.

(c) If W_1, W_2 are closed, disjoint, and G -invariant in X , then $\pi(W_1)$ and $\pi(W_2)$ are disjoint in Y .

We often write a good categorical quotient as $\pi: X \rightarrow X//G$. Here are some properties of good categorical quotients.

Theorem 5.0.6. Let $\pi : X \rightarrow X//G$ be a good categorical quotient. Then:

(a) Given any diagram



where ϕ is a morphism such that $\phi(g \cdot x) = \phi(x)$ for $g \in G$ and $x \in X$, there is a unique morphism $\bar{\phi}$ making the diagram commute, i.e., $\bar{\phi} \circ \pi = \phi$.

(b) π is surjective.

(c) A subset $U \subseteq X//G$ is open if and only if $\pi^{-1}(U) \subseteq X$ is open.

(d) If $U \subseteq X//G$ is open and nonempty, then $\pi|_{\pi^{-1}(U)} : \pi^{-1}(U) \rightarrow U$ is a good categorical quotient.

(e) Given points $x, y \in X$, we have

$$\pi(x) = \pi(y) \iff \overline{G \cdot x} \cap \overline{G \cdot y} \neq \emptyset.$$

From now on, suppose G is an affine algebraic group and G acts algebraically on X

If $\pi : X \rightarrow X//G$ is a good categorical quotient
Then

(a) If $p \in X//G$, then $\pi^{-1}(p)$ contains a unique closed G -orbit.

(b) π induces a bijection

$$\{\text{closed } G\text{-orbits in } X\} \simeq X//G.$$

III. Generic quotient

Proposition 5.0.8. Let $\pi : X \rightarrow X//G$ be a good categorical quotient. Then the following are equivalent:

(a) All G -orbits are closed in X .

(b) Given points $x, y \in X$, we have

$$\pi(x) = \pi(y) \iff x \text{ and } y \text{ lie in the same } G\text{-orbit.}$$

(c) π induces a bijection

$$\{G\text{-orbits in } X\} \simeq X//G.$$

(d) The image of the morphism $G \times X \rightarrow X \times X$ defined by $(g, x) \mapsto (g \cdot x, x)$ is the fiber product $X \times_{X//G} X$.

It is called geometric quotient if it satisfies the conditions of prop 5.0.8, denoted by $\pi: X \rightarrow X/G$

How to check the conditions in the definition of good categorical quotient?

A simple case: if $X = \text{Spec}(R)$ and G is a reductive group (e.g. finite group, tori, semisimple group such as $SL_n(\mathbb{C})$)

Proposition 5.0.9. Let G be a reductive group acting algebraically on an affine variety $X = \text{Spec}(R)$. Then:

- (a) R^G is a finitely generated $\mathbb{C}$ -algebra.
- (b) The morphism $\pi: X \rightarrow \text{Spec}(R^G)$ induced by $R^G \subseteq R$ is a good categorical quotient.

IV. Almost generic quotient

Proposition 5.0.11. Let $\pi: X \rightarrow X//G$ be a good categorical quotient. Then the following are equivalent:

- (a) X has a G -invariant Zariski dense open subset U_0 such that $G \cdot x$ is closed in X for all $x \in U_0$.
- (b) $X//G$ has a Zariski dense open subset U such that $\pi|_{\pi^{-1}(U)}: \pi^{-1}(U) \rightarrow U$ is a geometric quotient.

Example

Generic quotient

Example 5.0.1. Let $\mu_2 = \{\pm 1\}$ act on $\mathbb{C}^2 = \text{Spec}(\mathbb{C}[s, t])$, where $-1 \in \mu_2$ acts by multiplication by -1 . Note that every orbit consists of two elements, with the exception of the orbit of the origin, which is the unique fixed point of the action.

The ring of invariants $\mathbb{C}[s, t]^{\mu_2} = \mathbb{C}[s^2, st, t^2]$ is the coordinate ring of the affine toric variety $\mathbb{V}(xz - y^2)$. Hence we get a map

$$\Phi: \mathbb{C}^2/\mu_2 \longrightarrow \text{Spec}(\mathbb{C}[s, t]^{\mu_2}) = \mathbb{V}(xz - y^2) \subseteq \mathbb{C}^3$$

where the orbit $\mu_2 \cdot (a, b)$ maps to (a^2, ab, b^2) . This is easily seen to be a bijection, so that $\text{Spec}(\mathbb{C}[s, t]^{\mu_2})$ is the perfect way to make $\mathbb{C}^2/\mu_2$ into an affine variety.

This is actually an example of the toric morphism induced by changing the lattice—see Examples 1.3.17 and 1.3.19. $\diamond$

almost
generic
quotient

Example 5.0.2. Let $\mathbb{C}^*$ act on $\mathbb{C}^4 = \text{Spec}(\mathbb{C}[x_1, x_2, x_3, x_4])$, where $\lambda \in \mathbb{C}^*$ acts via

$$\lambda \cdot (a_1, a_2, a_3, a_4) = (\lambda a_1, \lambda a_2, \lambda^{-1} a_3, \lambda^{-1} a_4).$$

In this case, the ring of invariants is

$$\mathbb{C}[x_1, x_2, x_3, x_4]^{\mathbb{C}^*} = \mathbb{C}[x_1 x_3, x_2 x_4, x_1 x_4, x_2 x_3],$$

which gives the map

$$\Phi : \mathbb{C}^4 / \mathbb{C}^* \longrightarrow \text{Spec}(\mathbb{C}[x_1, x_2, x_3, x_4]^{\mathbb{C}^*}) = \mathbf{V}(xy - zw) \subseteq \mathbb{C}^4$$

where the orbit $\mathbb{C}^* \cdot (a_1, a_2, a_3, a_4)$ maps to $(a_1 a_3, a_2 a_4, a_1 a_4, a_2 a_3)$. Then we have (Exercise 5.0.2):

- Φ is surjective.
- If $p \in \mathbf{V}(xy - zw) \setminus \{0\}$, then $\Phi^{-1}(p)$ consists of a single $\mathbb{C}^*$ -orbit which is closed in $\mathbb{C}^4$.
- $\Phi^{-1}(0)$ consists of all $\mathbb{C}^*$ -orbits contained in $\mathbb{C}^2 \times \{(0, 0)\} \cup \{(0, 0)\} \times \mathbb{C}^2$. Thus $\Phi^{-1}(0)$ consists of infinitely many $\mathbb{C}^*$ -orbits.

This looks bad until we notice one further fact (Exercise 5.0.2):

- The fixed point $0 \in \mathbb{C}^4$ gives the unique closed orbit mapping to 0 under Φ .

If $(a, b) \neq (0, 0)$, then an example of a non-closed orbit is given by

$$\mathbb{C}^* \cdot (a, b, 0, 0) = \{(\lambda a, \lambda b, 0, 0) \mid \lambda \in \mathbb{C}^*\}$$

since $\lim_{\lambda \rightarrow 0} (\lambda a, \lambda b, 0, 0) = 0$. However, restricting to closed orbits gives

$$\{\text{closed } \mathbb{C}^* \text{-orbits}\} \simeq \mathbf{V}(xy - zw).$$

good
categorical
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Example 5.0.3. Let $\mathbb{C}^*$ act on $\mathbb{C}^{n+1} = \text{Spec}(\mathbb{C}[x_0, \dots, x_n])$ by scalar multiplication. Then the ring of invariants consists of polynomials satisfying

$$f(\lambda x_0, \dots, \lambda x_n) = f(x_0, \dots, x_n)$$

for all $\lambda \in \mathbb{C}^*$. Such polynomials must be constant, so that

$$\mathbb{C}[x_0, \dots, x_n]^{\mathbb{C}^*} = \mathbb{C}.$$

It follows that the “quotient” is $\text{Spec}(\mathbb{C})$, which is just a point. The reason for this is that the only closed orbit is the orbit of the fixed point $0 \in \mathbb{C}^{n+1}$. $\diamond$

useful references

I. Dolgachev, lectures on invariant theory

Quotient construction of toric varieties.

Theorem 5.1.11. Let X_Σ be a toric variety without torus factors and consider the toric morphism $\pi : \mathbb{C}^{\Sigma(1)} \setminus Z(\Sigma) \rightarrow X_\Sigma$ from Proposition 5.1.9. Then:

(a) π is an almost geometric quotient for the action of G on $\mathbb{C}^{\Sigma(1)} \setminus Z(\Sigma)$. Thus

$$X_\Sigma \simeq (\mathbb{C}^{\Sigma(1)} \setminus Z(\Sigma)) // G.$$

(b) π is a geometric quotient if and only if Σ is simplicial.

pf of (a) Consider $\pi_b : U_b \rightarrow U_b$

① prove that π_b is a good categorical quotient

since G is reductive, it suffices to show that $\mathbb{C}[U_b] \cong \mathbb{C}[U_b]^G$

② Prove that $\pi_b : U_b \rightarrow U_b$ is geometric quotient $\Leftrightarrow b$ is simplicial

③ $\exists b \in \Sigma$ s.t. b is simplicial and $U_b \supseteq T_v$ is dense open.

$\Rightarrow \pi$ is almost geometric quotient.

- prop 5.1.9 (1) $\mathbb{C}^{\Sigma(1)} \setminus Z(\Sigma)$ is a toric variety
- (2) The toric morphism $\pi : \mathbb{C}^{\Sigma(1)} \setminus Z(\Sigma) \rightarrow X_\Sigma$ is constant on G -orbits

Proof. For part (a), let $\tilde{\Sigma}_0$ be the fan consisting of $\text{Cone}(e_\rho \mid \rho \in \Sigma(1))$ and its faces. Note that $\tilde{\Sigma}$ is a subfan of $\tilde{\Sigma}_0$. Since $\tilde{\Sigma}_0$ is the fan of $\mathbb{C}^{\Sigma(1)}$, we get the toric variety of $\tilde{\Sigma}$ by taking $\mathbb{C}^{\Sigma(1)}$ and then removing the orbits corresponding to all cones in $\tilde{\Sigma}_0 \setminus \tilde{\Sigma}$. By the Orbit-Cone Correspondence (Theorem 3.2.6), this is equivalent to removing the orbit closures of the minimal elements of $\tilde{\Sigma}_0 \setminus \tilde{\Sigma}$. But these minimal elements are precisely the primitive collections $C \subseteq \Sigma(1)$. Since the corresponding orbit closure is $\mathbf{V}(x_\rho \mid \rho \in C)$, removing these orbit closures means removing

$$Z(\Sigma) = \bigcup_C \mathbf{V}(x_\rho \mid \rho \in C).$$

For part (b), define $\bar{\pi} : \mathbb{Z}^{\Sigma(1)} \rightarrow N$ by $\bar{\pi}(e_\rho) = u_\rho$. Since the minimal generators of $\sigma \in \Sigma$ are given by u_ρ , $\rho \in \sigma(1)$, we have $\bar{\pi}_{\mathbb{R}}(\tilde{\sigma}) = \sigma$ by the definition of $\tilde{\sigma}$. Hence $\bar{\pi}$ is compatible with respect to the fans $\tilde{\Sigma}$ and Σ .

The map of tori induced by $\bar{\pi}$ is the map $(\mathbb{C}^*)^{\Sigma(1)} \rightarrow T_N$ from the exact sequence (5.1.2) (you will check this in Exercise 5.1.3). Hence, if $g \in G \subseteq (\mathbb{C}^*)^{\Sigma(1)}$ and $x \in \mathbb{C}^{\Sigma(1)} \setminus Z(\Sigma)$, then

$$\pi(g \cdot x) = \pi(g) \cdot \pi(x) = \pi(x),$$

where the first equality holds by equivariance and the second holds since G is the kernel of $(\mathbb{C}^*)^{\Sigma(1)} \rightarrow T_N$. This proves part (c) of the proposition. $\square$

- The group G

$$0 \rightarrow M \rightarrow \mathbb{Z}^{\Sigma(1)} \rightarrow \text{cl}(X_\Sigma) \rightarrow 0$$

$$\leadsto 1 \rightarrow \underbrace{\text{Hom}_{\mathbb{Z}}(\text{cl}(X_\Sigma), \mathbb{C}^*)}_{\substack{\cong \\ G}} \rightarrow \text{Hom}_{\mathbb{Z}}(\mathbb{Z}^{\Sigma(1)}, \mathbb{C}) \cong (\mathbb{C}^*)^{\Sigma(1)}$$

$$\rightarrow \underbrace{\text{Hom}_{\mathbb{Z}}(M, \mathbb{C}^*)}_{\substack{\cong \\ T_N}} \rightarrow 1$$
- ① $\text{cl}(X_\Sigma)$ is the character group of G
- ② G reductive
- ③

Given a basis $e_1, \dots, e_n$ of M , we have

$$G = \{(t_\rho) \in (\mathbb{C}^*)^{\Sigma(1)} \mid \prod_\rho t_\rho^{\langle m, u_\rho \rangle} = 1 \text{ for all } m \in M\}$$

$$= \{(t_\rho) \in (\mathbb{C}^*)^{\Sigma(1)} \mid \prod_\rho t_\rho^{\langle e_i, u_\rho \rangle} = 1 \text{ for } 1 \leq i \leq n\}.$$

- $S = \mathbb{C}[X_\rho \mid \rho \in \Sigma(1)]$ total coordinate ring of X_Σ
- $\text{Spec}(S) = \mathbb{C}^{\Sigma(1)}$

- $B(\Sigma)$, $Z(\Sigma) = V(B(\Sigma)) \subseteq \mathbb{C}^{\Sigma(1)}$

For each cone $\sigma \in \Sigma$, define the monomial

$$x^{\hat{\sigma}} = \prod_{\rho \notin \sigma(1)} x_{\rho}.$$

Thus $x^{\hat{\sigma}}$ is the product of the variables corresponding to rays not in σ . Then define the *irrelevant ideal*

$$B(\Sigma) = \langle x^{\hat{\sigma}} \mid \sigma \in \Sigma \rangle \subseteq S.$$

A useful observation is that $x^{\hat{\tau}}$ is a multiple of $x^{\hat{\sigma}}$ whenever $\tau \preceq \sigma$. Hence, if $\Sigma_{\max}$ is the set of maximal cones of Σ , then

$$B(\Sigma) = \langle x^{\hat{\sigma}} \mid \sigma \in \Sigma_{\max} \rangle.$$

Definition 5.1.5. A subset $C \subseteq \Sigma(1)$ is a ***primitive collection*** if:

- (a) $C \not\subseteq \sigma(1)$ for all $\sigma \in \Sigma$.
- (b) For every proper subset $C' \subsetneq C$, there is $\sigma \in \Sigma$ with $C' \subseteq \sigma(1)$.

Proposition 5.1.6. The $Z(\Sigma)$ as a union of irreducible components is given by

$$Z(\Sigma) = \bigcup_C \mathbf{V}(x_{\rho} \mid \rho \in C),$$

where the union is over all primitive collections $C \subseteq \Sigma(1)$.